Contact

www.linkedin.com/in/krishnayalamanchili (LinkedIn)
krishnayalamanchili.blogspot.com (Personal)

Top Skills

Full-Stack Development
Leadership
Agentic Automation

Languages

German (Elementary)
Hindi (Limited Working)
Telugu (Native or Bilingual)
English (Full Professional)

Certifications

Learning Data Governance
Privacy by Design: Data Sharing

Krishna Yalamanchili

Principal AI Engineer @ Lantern UAS | AI Agents, Computer Vision
Fremont, California, United States

Summary

I’m a Principal AI/ML and Search Engineering leader with 16+ years of experience building intelligent systems, scalable data platforms, and enterprise AI solutions across Supply Chain Intelligence, EdTech, HRTech, Marketplace Platforms, and Physical-World AI systems.My work spans production-scale search and retrieval systems, multimodal AI pipelines, agentic architectures, distributed infrastructure, and enterprise-grade ML platforms. I specialize in transforming complex AI/ML concepts into scalable systems that drive measurable business impact.Over the course of my career, I have led and managed engineering and ML teams ranging from 5–25 engineers across startups and enterprise environments — balancing hands-on technical execution with architecture leadership, mentoring, and delivery ownership. ’ :• Enterprise-scale search, retrieval, and ranking systems processing billions of records and events• AI-powered enrichment pipelines for supply chain and entity resolution systems• Agentic AI workflows combining LLMs, retrieval systems, orchestration layers, evaluation frameworks, and multimodal intelligence• Large-scale distributed data and ML platforms using Spark, Kafka, OpenSearch, Kubernetes, and cloud-native architectures• Production systems involving semantic search, vector retrieval, reranking, NLP, computer vision, and real-time analytics :• Low-latency retrieval and RAG architectures• AI evaluation and observability frameworks• Multimodal AI systems across text, image, video, and structured data• GPU-aware inference optimization pipelines• AI governance and human-in-the-loop workflows• Detection, behavioral analysis, and real-time decision systemsI enjoy solving technically challenging problems, architecting systems from first principles, and helping organizations operationalize AI beyond experimentation into production-grade systems. :Generative AI | Agentic AI | LLMs | NLP | Search & Retrieval | RAG | Semantic Search | Multimodal AI | Computer Vision | Distributed Systems | MLOps | AI Evaluation & Observability | OpenSearch | Spark | Kafka | Kubernetes:Supply Chain Intelligence | B2B SaaS | EdTech | HRTech | Marketplace Platforms | Enterprise

AI | Physical-World AI | Security & Detection SystemsI’m particularly interested in building reliable AI systems at scale involving retrieval, multimodal intelligence, and agentic orchestration.

Experience

Lantern UAS
Principal AI Engineer
January 2026 - Present (6 months)

• Own product and engineering direction for PURPLE — a Predictive UAS Response Platform for Layered Engagement — including investor demo setup, roadmap construction, phase-level deployment/delivery plans, development setup guidelines, knowledge-transfer documentation, hiring support, team formation, and mentorship while evolving the platform from proof of concept to production-ready counter-UAS workflows.

• Built the real-time drone-detection pipeline: assembled a 200K curated drone dataset, fine-tuned YOLO-family detectors, evaluated RT-DETR-style approaches, and integrated video upload → inference → ByteSORT tracking → zone-centric, sensor-anchored localization, generating deterministic / replayable track timelines with tactical HUD overlays.

• Designed a Gemini-native multimodal agentic pipeline with six Layer-1 modality agents, Layer-2 environmental / signal fusion agents, and a Layer-3 ROE-grounded threat-triage agent; enforced strict Pydantic I/O contracts, prompt versioning, real-time pipeline telemetry, deterministic threat floors, and post-LLM safety-net escalation for zone-centric, sensor-anchored threat assessment.

• Implemented the policy and decision-support engine for zone SOPs, ROE rules, jurisdiction catalogs, action catalogs, escalation ladders, approval workflows, email notifications, execution outcomes, case-history snapshots, and evidence workflows; deploying across AWS and Azure environments and supporting H100 GPU-based detection / classification for ultra-low-latency requirements as well as CPU-centric deployments.

Altana
Principal Machine Learning Engineer
May 2024 - January 2026 (1 year 9 months)
San Francisco Bay Area

• Next-gen facility matcher: Initially built a hybrid BERT + LLM reranker that fused name, address, HS-code, graph-proximity, and shipment features to lift facility-matching precision/recall 13–15 %, then superseded it with a lean in-house BERT cross-encoder trained on 1.2 M curated pairs, nudging automatic matching recall to 90% without precision loss and delivering a 32× semantic-matching speed-up (8 s → 0.25s). Optimized OpenSearch queries with custom English and non-English analyzers and country detection.

• Agentic shipment enrichment: Designed a LangGraph / ReAct pipeline that plans, executes, and evaluates RAG calls to enrich importer/exporter data and goods descriptions with HS-level attributes—groundwork for automated labeling and multi-stage HS-code classification.

• Facility discovery & deduplication: Replaced the legacy matcher with a fine-tuned BERT vector index plus OpenSearch rules, boosting recall and enabling facility-grouping logic that reveals corporate hierarchies and removes duplicates.

• Architecture & standards leadership: Adopted the full Model Context Protocol (roles, function calls, long contexts) ahead of industry rollout and now spearhead an in-house BERT search model and multi-stage HS classifier to power next-gen supply-chain intelligence.

Learneo

Staff Software Engineer, Search Platform

March 2022 - May 2024 (2 years 3 months)

Redwood City, California, United States

• Search infrastructure: Migrated Elasticsearch → AWS OpenSearch and deployed a 1 B-vector FAISS + HNSW index (<20 ms latency), slashing infra costs 6× and laying the groundwork for conversational search.

• Relevance & discovery: Expanded query-suggestion index
270 K → 400 M terms with advanced n-gram analysis and A/B tuning, lifting click-through 60 %; added subject/concept classifiers for richer, context-aware results.

• Real-time & AI UX: Built Kafka/ksqlDB event-driven pipelines and high-throughput APIs, and authored GPT-3.5/4 prompt flows that power interactive chat experiences.

Walmart Global Tech

5 years 3 months

Staff Software Engineer, Facets Management, Catalog & Search

June 2019 - March 2022 (2 years 10 months)

Sunnyvale, California, United States

Patents:

https://patents.google.com/patent/US20200175553A1/en

https://patents.google.com/patent/US20210240700A1/en

• Product discovery leadership: Invented and patented (US 20200175553 / US 20210240700) dynamic facet and deal-curation methods; led the Facets Management and Holiday Deals Shelving platforms that power Walmart.com search and merchandising.

• Platform scale & reliability: Engineered rule-based, multi-tenant facet pipelines with Apache Storm, Kafka, and Elasticsearch—adding versioning, metrics, and exhaustive logging—to support Black-Friday-level traffic with seamless monitoring and auditing.

• End-to-end delivery: Built and shipped Java/Scala microservices and React/Redux UIs backed by GraphQL, Cassandra, and MongoDB, unifying content, search, and promotions for millions of daily users.

Senior Software Engineer, Content Quality Scoring, Catalog & Search

January 2017 - May 2019 (2 years 5 months)

Sunnyvale, CA

• Real-time product-quality scoring: Built a reactive pipeline that computes attribute and confidence scores on millions of SKUs, raising data quality for Walmart.com search and merchandising.

• Big-data stream processing: Architected Java/Scala pipelines with Kafka, Spark, Storm, and Akka to aggregate and filter massive JSON records in real time, enabling instant analytics and feedback loops.

• High-throughput data delivery: Orchestrated Akka Streams and Storm topologies to push processed data at scale into Elasticsearch, Cassandra, and HDFS, ensuring low-latency availability for downstream systems.

Cisco

Senior Software Engineer, Marketing & User Acquisition

April 2016 - January 2017 (10 months)
Richardson, TX

• Developed front-end and back-end systems using AngularJS, Java, Spring, Postgres, Cassandra, and Redis.
• Owned and managed Email and Marketing components for Cisco Spark, overseeing implementation of security measures and executing various AB testing and marketing campaigns using Test and Target and other Adobe Marketing Cloud features.
• Wrote code for the creation of multi-threaded and event-driven applications, both standalone and those interacting with servers or services.

TMX Finance
Senior Java Software Engineer, Web Developer
February 2015 - April 2016 (1 year 3 months)

• Created Java based web services for TMX Finance's internal software tool.
• Crafted HTML5-based UI pages with Angular.js, complemented by backend services developed with Spring Boot. Implemented a dynamic event-driven UI using a combination of Oracle triggers, Apache ActiveMQ message queues, JMS, and the Spring WebSockets framework.

Hewlett-Packard
4 years 5 months

Senior Systems/Software Engineer, HP StoreAll(IBRIX)
April 2011 - January 2015 (3 years 10 months)
Andover, MA

• Developed innovative features in JAVA & Scala for our cutting-edge plugin-based architecture. This involved creating a JAX-RS and Jersey based REST API utilizing the Grizzly server for an asynchronous distributed system. The development adhered to Test Driven Development principles, employing JUnit and TestNG.
• Spearheaded the creation of a REST API and CLI for FSCK, along with various other StoreAll file system services.
• Took ownership of hardware registration, monitoring, and hardware health reporting components within the Administrative System software, which operates across all HP StoreAll products.

Software Designer, HP Storage Essentials
September 2010 - April 2011 (8 months)
Andover, MA

• Contributed to the development of a Save & Restore feature in JAVA and PL/SQL, playing a crucial role in preventing data loss in the event of a database corruption.
• Established a new HP Storage Works 4400 Enterprise Virtual Array, integrating it with VMware ESX software and VCenter for optimal performance.

EBS Inc
Software Engineer
January 2010 - September 2010 (9 months)
Groton, MA

• Utilized C programming to implement key features for EBS's embedded FAT-based file system, embedded web browser, and CIFS and UPnP products.

The MathWorks
Software Engineer Intern, EDA Targets
January 2009 - August 2009 (8 months)
Natick, MA

• Engineered GNU Makefiles, C, and Matlab code to introduce functionalities for the EDA Targets Tool. This tool establishes a connection between Matlab and Simulink with EDA tools such as Mentor Graphics ModelSim, Cadence Incisive, and Synopsys Discovery.

Education

Northeastern University
Master's Degree, Computer Engineering · (2007 - 2009)

Jawaharlal Nehru Technological University
Bachelor's Degree, Electrical and Electronics Engineering · (2003 - 2007)